DATA ANALYST PORTFOLIO PROJECT · BUSINESS RETROSPECTIVE

The Data I Had, But Didn’t Know How to Use

I ran my own business, generated leads and watched ad costs rise while too few leads became customers. Later I used SQL, Python, BigQuery and Looker Studio to find out what I missed.

Sun Sahar

Meta Ads · Jun 2024 – Jan 2026 · ₪39,487 spend · 1,923 leads

THE BUSINESS

I steered by cost per lead

A trading-education business. A free course brought people in; a phone call was meant to sell the full course with mentorship.

Meta ads kept the leads coming, but costs rose and few leads reached a sales call. I changed creatives, budgets, prices and the sales process — and judged it mainly by two numbers: how many leads, and what each one cost.

What could my own data have told me?

₪39,487
Meta ad spend, Jun 2024 – Jan 2026

1,923 leads
at ₪15.28 each — counted by Meta, not customers

38 paying customers
₪142,234 paid in total · I never recorded how each one found me

WHY LEADS GOT MORE EXPENSIVE

Most of the rise happened before the click

Cost per lead, Feb → Jun 2025: ₪8.78 → ₪17.74. Where the increase came from:



Cost per lead

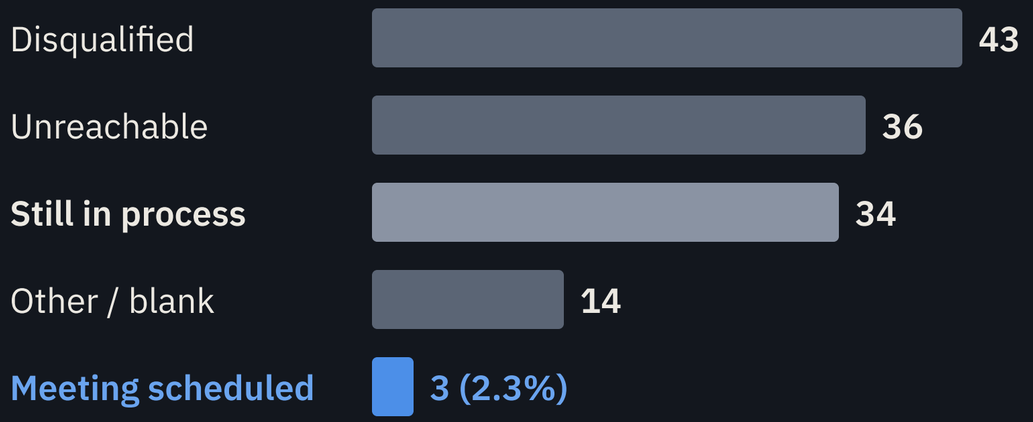
Feb ₪8.78
Mar ₪17.76
Apr ₪12.72
May ₪16.51
Jun ₪17.74
Jul ₪10.25

What it means: I blamed the form, but more clickers signed up. The problem came earlier: ads cost more to show and fewer people clicked. The data can't say why, so I'd start by testing new ads and audiences.

FROM LEADS TO SALES

The Missing Link: From Leads to Sales

130 leads in my CRM, checked once (29 Apr – 20 May 2025)



A one-time check: 34 leads were still open and may have moved on later. "Scheduled" doesn't mean the meeting happened. It points to a weak step after the lead, but doesn't prove why.

38 paying customers

in my receipts. They came through word of mouth, ads, webinars and phone calls — and I never recorded which one brought whom.

≈ ₪715 a rough estimate

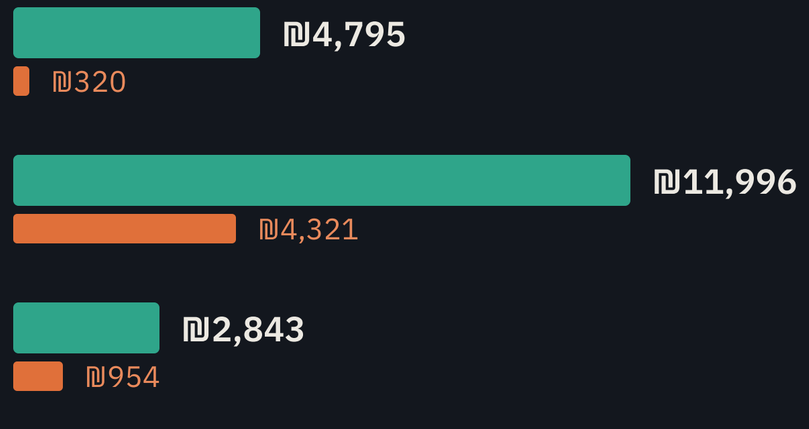
in ad cost per scheduled meeting, if this sample were typical (130 leads × ₪16.51 ÷ 3)

The landing page cost 2.4× per lead — but it asked for a meeting, not a free course.

WHERE THE MONEY CAME FROM

Payments rose 2.5× while the lead campaigns ran

Per month: ■ customer payments ■ Meta ad spend



Before the lead campaigns, ₪38,360 came in with only ₪2,560 spent on ads — word of mouth and calls were already selling. And during the campaigns we also tried to call every lead.

A simple comparison

≈ 11 of 38 customers

Payments from about 11 average customers (₪3,743 each) equal the whole ₪39,487 ad spend. This is money in, not profit: course and sales costs are not included.

Not proof

No payment shows which channel brought the customer, so I can't say the ads caused these sales. The timing fits, nothing more.

WHAT I COULD NOT SEE

Three blind spots and a missed warning

No lead was followed to a sale

Calls, meetings and payments were never connected to the ads, so I couldn't judge any campaign by sales.

Many ads, one name

Different ads ran under one name, so for 60% of the spend I can't tell which ad worked.

Tests judged by the wrong number

₪3,501 spent testing new offers was measured by clicks, chats and views, not by sales.

TESTING A RULE ON MY OWN HISTORY

1 March 2025

A simple daily alert — cost per lead in the last 7 days 30% above the days before — would have warned me about a month before my March report.

While the alert was on: ₪8,874 bought 512 leads at ₪17.33.

Tested on past data only. 30% is my choice, not a proven number. An alert means “take a look”, not “stop”.

06 · Queries A2 (alert test), A3, A5 · my own account of the ads and offers

WHAT-IF · A PLAN, NOT A FORECAST

What-If: Rebuilding My Next Quarter

If I could plan the quarter after July 2025 again: a ₪13,000 budget (my average monthly ad spend × 3), using cost-per-lead numbers from real months.

Scenario	Cost per lead	Leads	Meetings at 2.3% · today	At 5% · goal
Good month	₪12.72	1,022	24	51
Base	**₪14.07**	**924**	**21**	**46**
Bad month	₪17.76	732	17	37

The number to track: cost per scheduled meeting

₪610 → ₪281

ad cost only, base case, if the 5% goal is reached

Meetings matter more than cheaper leads

37 vs 24 meetings

worst ad month at 5% vs best ad month at 2.3%

WHAT-IF · THE PLAN I WOULD RUN

Measure first, test, decide by rule

Week 0 · Measure

- A unique name for every ad
- Every lead saved with the ad it came from
- Track each lead: called, meeting scheduled, meeting held, sale, amount
- Every customer tagged by source: ad, webinar, referral or call

After one quarter: cost per sale can finally be measured

Three tests

- **Offer:** free course vs fit meeting, same account and audience, 50/50 for 3 weeks
- **A screening question** in the form: worth it if meetings rise more than cost per lead
- **Ads:** three named ads, replaced on a fixed schedule

All judged by cost per scheduled meeting

Decision rules

- **Daily:** cost per lead 30% above normal for 3 days → check the ads and audience
- **Weekly:** cost per meeting above ₪600 for 2 weeks → rethink where the budget goes
- **Capacity:** about 300 leads a month, each called within 24 hours

Starting numbers, to be checked on new data

WHAT I LEARNED

I was measuring the easy step

1

Measure the step that pays, not the one that is easy to count.

2

Cost per lead tells you about the ads, not about the business.

3

Record where every customer came from — and what every test sold.

What this can't show: which channel brought each customer (so no CAC or ROAS), or profit. Leads are not customers, and the meeting rate comes from one CRM check.

Live dashboard: datastudio.google.com/reporting/517f4b3d-41fb-49b6-b738-0b5f93a7f9a8
Case studies, SQL and code: github.com/Ssahar6/python

APPENDIX A · HOW I DID IT

Every number checked before it was used

Data

- Meta Ads daily exports, 2 accounts
- 92,028 rows; 99% were export padding — only 595 ad-days with spend count
- Student payments: 50 receipts, 38 customers, names replaced by IDs
- CRM evidence as aggregates only

Tools

- **Python:** cleaning the data, removing names, checks
- **BigQuery SQL:** CTEs, window functions, rolling back-test
- **Looker Studio:** dashboard live on BigQuery
- **AI-assisted:** AI tools wrote the code; I set the questions, ran the queries and interpreted the results

Validation

- Key ad totals matched between Python and BigQuery
- Payments checked: 50 payments, 38 customers, ₪142,234
- Rates calculated from totals, not by averaging daily rates

Case studies, methodology, SQL and code: github.com/Ssahar6/python

APPENDIX B · MARKETING METRICS

Definitions and my values

Metric	Formula	Instant Form	Landing page	Note
CPM	spend ÷ impressions × 1,000	₪48.84	₪69.46	cost per 1,000 impressions
CTR	link clicks ÷ impressions	0.80%	0.94%	ad appeal
CPC	spend ÷ link clicks	₪6.08	₪7.39	cost per link click
LPV	page loads after a click	no page	714 of 755	95% arrived
CVR	leads ÷ link clicks	43.2%	30.6%	sign-up rate
CPL	spend ÷ Meta-attributed leads	₪14.07	₪24.16	₪15.28 overall
CAC	spend ÷ new paying customers	n/a	n/a	no sales link
ROAS	revenue ÷ ad spend	n/a	n/a	revenue not linked to ads

```
CPL = (CPM ÷ 1,000) ÷ CTR ÷ CVR
```

CTR and CVR are shown as percentages; in the formula they are decimals (0.80% = 0.0080, 43.2% = 0.432).

B · All active days of each mechanism · BigQuery queries A1, M1, Q006 · the two mechanisms are different offers in different accounts